Majority composition in randomised query complexity with expected small-error inner cost

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Editorial scope. Counted once: complete majority-lemma result, incorporating general GapOr reduction and every source-local leaf/abort/minimax support argument. Worst-case R and expected bar-R costs, partial domains and all error conventions remain distinct as authored. No separate count for hard sides, parameter ranges, GapOr reduction or support lemmas.

Allowed standard prerequisites (editorial).

- Partial-function decision trees, conical juntas, imported direct-sum/nonnegative GapOr bounds and abort-tree minimax. Exact original definitions and imported [5, Theorem4], [7] direct-sum and Lemmas8/9 retained. All source-local error reduction, Simulation, Leaf, HardSide and Multileaf proofs included. Full25 published references. Imported independent-paper proofs not recursively reproduced.

Author result and reduction

We assume $n \geq 3$ for simplicity of notation throughout the paper.

Theorem2 (Maj lemma). $R(\mathrm{Maj} \circ g^n) = \Theta(n \cdot \overline{R}_{1/n}(g))$ for every partial function g.

Previously, Ben-David et al. [5] proved Theorem2 in the special case $g = \mathrm{GapOr}$. Here $\mathrm{GapOr} = \mathrm{GapOr}_m$ is the m-bit partial function defined by $\mathrm{GapOr}(x) = \mathrm{Or}(x)$ on inputs of Hamming weight $|x| \in \{0, m/2\}$ and is undefined otherwise. This is a particularly clean example of a function whose query complexity behaves as (assuming $m \geq \log(1/\epsilon)$)

$$\overline{R}_\epsilon(\mathrm{GapOr}) = \Theta(\log(1/\epsilon)).$$

We prove Theorem2 by a direct reduction to this previous result! Our more general result says, informally, that error reduction is necessary for any composed function $f \circ g^n$ if it is necessary in the special case $g = \mathrm{GapOr}$. Our key conceptual insight is to formulate a sense in which every g can be "simulated" by GapOr. There is, however, a slight technical caveat. For the reduction to work, we need to assume that the lower bound for $f \circ \mathrm{GapOr}^n$ holds not only against randomised decision trees but also against a more powerful model

called ϵ-approximate nonnegative degree $\deg^+_\epsilon$ (aka conical junta degree, partition bound), which we will recall in Section2.

Theorem3 (Reduction to GapOr). If a function f satisfies $\deg^+_\epsilon(f \circ h^n) \geq \Omega(n \log n)$ for some constant $\epsilon > 0$ and for both $h \in \{\mathrm{GapOr}_{\log n}, \neg\mathrm{GapOr}_{\log n}\}$, then

$$\forall g: \quad R(f \circ g^n) = \Omega(n \cdot \overline{R}_{1/n}(g)).$$

Theorem2 follows immediately by combining Theorem3 with [5, Theorem4], which proved the required nonnegative degree lower bound for $\mathrm{Maj} \circ \mathrm{GapOr}^n$ (we only note that their proof works equally well for $\neg$GapOr in place of GapOr). In fact, the nonnegative degree lower bound holds more generally for any $(2n+1)$-bit outer function that agrees with Maj on inputs of weight n and $n+1$. For example, Xor is such a function, and hence the Xor lemma of Brody et al. [8] can be recovered using Theorem3. However, the original proof in [8] is much simpler than ours, and moreover, the result of [8] actually characterises $\overline{R}_\epsilon(\mathrm{Xor} \circ g^n)$ for all $\epsilon > 0$ while we focus on the bounded-error case $\epsilon = 1/3$. Our goal for the rest of the paper is to prove Theorem3. For randomised query complexity, a complete answer [to the direct sum problem] was recently obtained by Blais and Brody [7] (improving on [17,6]). Their result states

$$\forall g: \quad R(g^n) = \Theta(n \cdot \overline{R}_{1/n}(g)). \qquad \text{(Direct sum)}$$

2 Query complexity basics

We study partial boolean functions $f : \{0,1\}^n \to \{0,1,*\}$. The domain of the function is $\mathrm{dom}(f) := f^{-1}(\{0,1\})$ and the inputs $f^{-1}(*)$ are undefined. We say f is total if $\mathrm{dom}(f) = \{0,1\}^n$. For partial functions f and g, their composition $f \circ g^n$ is defined by $(f \circ g^n)(x_1, \ldots, x_n) := f(g(x_1), \ldots, g(x_n))$ if $x_i \in \mathrm{dom}(g)$ for all $i \in [n]$; otherwise $(f \circ g^n)(x_1, \ldots, x_n) := *$. Standard references for boolean function complexity are [9,18].

Decision trees. A (deterministic) decision tree t is an algorithm for computing a boolean function on an unknown input $x \in \{0,1\}^n$. The algorithm repeatedly queries the input variables $x_i \in \{0,1\}$ in some order (which can depend on outcomes of queries made so far) until eventually producing an output $t(x)$. Such an algorithm can be represented as a binary tree, with internal nodes labelled with variables x_i, outgoing edges of the internal nodes labelled with query outcomes ($x_i = 0$ and $x_i = 1$), and leaves labelled with output values. Each input x determines a unique root-to-leaf path, obtained by following the query outcomes consistent with x. The most important cost measure of t is its depth, denoted $\mathrm{depth}(t)$, which is the longest root-to-leaf path in the tree and equals $\max_x q(t,x)$ where $q(t,x)$ denotes the number of queries made by t on input x. A randomised decision tree T is a distribution over deterministic decision trees $t \sim T$. We say T computes $f : \{0,1\}^n \to \{0,1,*\}$ with error ϵ if for every $x \in \mathrm{dom}(f)$ we have $\Pr_{t \sim T}[t(x) = f(x)] \geq 1 - \epsilon$. There are two cost measures for T: the (worst-case) depth is the maximum depth of any decision tree in the support of T; the expected depth is $\max_x \mathbb{E}_{t \sim T}[q(t,x)]$. The ϵ-error query complexity of f, denoted $R_\epsilon(f)$, is the least depth of a randomised decision tree that computes f with error ϵ. The ϵ-error expected query complexity, denoted $\overline{R}_\epsilon(f)$, is defined analogously.

Error reduction. It is well known that the error probability of an algorithm (computing a boolean-valued function) can be reduced from any constant $1/2-\delta$, where $\delta > 0$, to any other constant $\epsilon > 0$ by repeating the algorithm constantly many times (in fact, $O(\log(1/\epsilon)/\delta^2)$ many) and outputting the majority answer. Hence we often set $\epsilon := 1/3$ and omit ϵ from notation. In this bounded-error regime, we have $\overline{R}(f) \leq R(f) \leq O(\overline{R}(f))$ where the second inequality follows by truncating executions that query many more bits than the expectation. For vanishing $\epsilon = o(1)$ (as $n \to \infty$), it is possible that $\overline{R}_\epsilon(f) \leq o(R_\epsilon(f))$. For example, consider the partial $2n$-bit function f where the task is to distinguish inputs of the form $x0^n$ from inputs of the form $0^n x$ with the promise that $|x| = n/2$. We have $\overline{R}_{1/n}(f) = O(1)$ while $R_{1/n}(f) = \Theta(\log n)$. In this small-error regime, the following fine-grained error reduction calculation will be useful.

Claim7. $\overline{R}_{\epsilon^k}(f) \leq 4k \cdot \overline{R}_\epsilon(f)$ for every $k \geq 1$ and $\epsilon \leq 1/16$.

Proof. Suppose T computes f with error ϵ and consider the algorithm T' that runs T $4k-1$ times and outputs the majority answer. Then T' errs iff at least $2k$ of the runs err. This happens with probability at most $\sum_{i=2k}^{4k-1} \binom{4k-1}{i} \epsilon^i (1-\epsilon)^{4k-1-i} \leq 2^{4k}\epsilon^{2k} \leq \epsilon^k$. □

Leaf indicators. Let t be a decision tree with n-bit inputs. We denote by $L(t)$ the set of its leaves and by $\ell_x^t \in L(t)$ the unique leaf reached on input x. We often identify a leaf $\ell \in L(t)$ with its associated leaf indicator function $\ell : \{0,1\}^n \to \{0,1\}$ defined by $\ell(x) := 1$ iff input x reaches leaf ℓ. Thus each ℓ is simply a conjunction of at most depth(t) literals (x_i or $\overline{x}_i$) determined by the unique root-to-ℓ path in t. If t outputs boolean values, we let $A(t) \subseteq L(t)$ denote the set of accepting leaves, that is, those that output 1. Since the leaf indicators have pairwise disjoint supports, we can write the function computed by t as

$$t(x) = \sum_{\ell \in A(t)} \ell(x). \tag{1}$$

Nonnegative degree. Let $p : \{0,1\}^n \to \mathbb{R}_{\geq 0}$ be a nonnegative function. We say p is a nonnegative d-junta if it depends on at most d of its variables. For example, if t is a depth-d decision tree, then each $\ell \in L(t)$ is a nonnegative d-junta. More generally, we say that p is a conical junta of degree d if it can be written as a conical combination of nonnegative d-juntas, that is, $p(x) = \sum_i a_i q_i(x)$ where $a_i \geq 0$ are nonnegative scalars and the q_i are nonnegative d-juntas. For example, the function computed by t is a degree-d conical junta, as given by the expression(1). The nonnegative degree of p, denoted $\deg^+(p)$, is the least d such that p is a degree-d conical junta. Let $f : \{0,1\}^n \to \{0,1,*\}$ be a partial function. We say that p ϵ-approximates f if $p(x) \in f(x) \pm \epsilon$ for every $x \in \mathrm{dom}(f)$. The ϵ-approximate nonnegative degree of f, denoted $\deg_\epsilon^+(f)$, is the least degree of a conical junta that ϵ-approximates f. For example, if T is a depth-d randomised ϵ-error decision tree for f, then there exists a degree-d conical junta p_T that ϵ-approximates f, namely,

$$p_T(x) := \mathbb{E}_{t \sim T}[t(x)] \in f(x) \pm \epsilon.$$

This shows that $\deg_\epsilon^+(f) \leq R_\epsilon(f)$. The gap betweeen $\deg_{1/3}^+(f)$ and $R(f)$ can be huge for partial functions. For example, consider the n-bit UniqueOr

defined by $\mathrm{UniqueOr}(x) = \mathrm{Or}(x)$ for inputs of weight $|x| \in \{0,1\}$ and undefined othwerwise. Then $\deg^+(\mathrm{UniqueOr}) = 1$ (computed by $\sum_i x_i$) while $R(\mathrm{UniqueOr}) = \Theta(n)$. For total functions, the gap is at most polynomial [9]. Nonnegative degree has been studied under many names: (one-sided) partition bound [16], WAPP query complexity [14,5], and query complexity "in expectation" [19].

Necessary Leaf and Multileaf Lemma definitions (Section3.1)

For a leaf $\ell \in L(t)$ and any input distribution μ we write for short $\ell(\mu) := \mathbb{E}_{x\sim\mu}[\ell(x)] = \Pr_{x\sim\mu}[\ell(x) = 1]$. Fix a partial function $g : \{0,1\}^m \to \{0,1,*\}$. Let $\mu = \frac{1}{2}(\mu^0 + \mu^1)$ be a balanced distribution where μ^i is supported on $g^{-1}(i)$. For a leaf ℓ over m bits, a "hard side" $b \in \{0,1\}$, and an error parameter $\epsilon \geq 0$, we define

$$\ell \text{ is } (\epsilon,\mu,b)\text{-noisy} \quad \iff \quad \ell(\mu^{1-b}) \geq \epsilon \cdot \ell(\mu^b).$$

For simplicity of notation, for small quantities $a, b \in [0,1]$, we write $a \ll b$ (resp. $a \lll b$) to mean $a \leq cb$ (resp. $a^c \leq b$) for a sufficiently small constant $c > 0$.

Leaf Lemma. For every partial g and $0 < \epsilon \lll \delta \ll 1$, there exists a distribution $\mu = \frac{1}{2}(\mu^0 + \mu^1)$ over $\mathrm{dom}(g)$ and a hard side $b \in \{0,1\}$ such that for every deterministic tree t and $i \in \{0,1\}$:

$$\frac{\mathbb{E}_{x\sim\mu^i}[q(t,x)]}{\overline{R}_\epsilon(g)} \lll \delta \quad \Longrightarrow \quad \Pr_{x\sim\mu^i}[\ell_x^t \text{ is } (\epsilon,\mu,b)\text{-noisy}] \geq 1-\delta.$$

In order to use Leaf Lemma in the context of composed functions, we generalise it to the direct sum setting where the inputs come from $\mathrm{dom}(g^n) := \mathrm{dom}(g)^n$. Let ℓ be a leaf over nm bits and write $\ell(x) = \prod_{i\in[n]} \ell_i(x_i)$ where $x_i \in \{0,1\}^m$ and each ℓ_i is over m bits. We define

$$\ell \text{ is } (\delta,\epsilon,\mu,b)\text{-noisy} \quad \iff \quad \ell_i \text{ is } (\epsilon,\mu,b)\text{-noisy for at least } (1-\delta)n \text{ many } i \in [n].$$

Our generalised lemma says that we will typically reach a noisy leaf if we run a shallow decision tree on a random input from the product distribution $\mu^y := \mu^{y_1} \times \cdots \times \mu^{y_n}$ where $y \in \{0,1\}^n$.

Multileaf Lemma. For every partial g and $0 < \epsilon \lll \delta \ll 1$, there exists a distribution $\mu = \frac{1}{2}(\mu^0 + \mu^1)$ over $\mathrm{dom}(g)$ and a hard side $b \in \{0,1\}$ such that for every deterministic tree t taking inputs from $\mathrm{dom}(g^n)$ and having $\mathrm{depth}(t)/(n\overline{R}_\epsilon(g)) \lll \delta$,

$$\forall y \in \{0,1\}^n : \quad \Pr_{x\sim\mu^y}[\ell_x^t \text{ is } (\delta,\epsilon,\mu,b)\text{-noisy}] \geq 1-\delta.$$

3.2 Proof of Theorem3

We conclude this overview section with a proof of Theorem3 using Multileaf Lemma. We start with a lemma that shows how the noisy leaves in the domain

of g^n can be "simulated" by low-degree conical juntas in the domain of GapOr^n. For simplicity, we state the lemma assuming a hard side $b = 0$; an analogous lemma holds for $b = 1$ by replacing GapOr with $\neg$GapOr.

Simulation Lemma. Let ℓ be a $(\delta, \epsilon, \mu, 0)$-noisy leaf over the variables of g^n. There exists a conical junta $p_\ell : (\{0,1\}^{\log n})^n \to \mathbb{R}_{\geq 0}$ of degree at most $n \cdot [\delta \log n + \log(1/\epsilon)]$ such that

$$\forall x \in \mathrm{dom}(\mathrm{GapOr}^n_{\log n}) : \quad p_\ell(x) = \ell(\mu^{\mathrm{GapOr}^n_{\log n}(x)}).$$

Proof. We start by defining three conical juntas in the domain of GapOr_m for $m := \log n$. Let S^m_k be the distribution over multisets obtained by picking k random elements from $[m]$ with replacement.

$$\begin{aligned}
q_1(x) &:= \tfrac{2}{m} \sum_{i \in [m]} x_i && \text{of degree } 1,\\
q_2(x) &:= \prod_{i \in [m]} \overline{x}_i && \text{of degree } m = \log n,\\
q_3(x) &:= \mathbb{E}_{S \sim S^m_k} \prod_{i \in S} \overline{x}_i && \text{of degree } k := \log(1/\epsilon).
\end{aligned}$$

Note the following output values:

$$\begin{aligned}
\forall x \in (\mathrm{GapOr}_m)^{-1}(0) : &\quad q_1(x) = 0, \quad q_2(x) = 1, \quad q_3(x) = 1,\\
\forall x \in (\mathrm{GapOr}_m)^{-1}(1) : &\quad q_1(x) = 1, \quad q_2(x) = 0, \quad q_3(x) = 2^{-k} = \epsilon.
\end{aligned}$$

Let $y = (y^1, \ldots, y^n)$ be the input variables of g^n. We write $\ell(y) = \prod_i \ell_i(y^i)$ so that $\ell(\mu^{\mathrm{GapOr}^n_m(x)}) = \prod_i \ell_i(\mu^{\mathrm{GapOr}_m(x^i)})$. We simulate each factor in this product separately. For $i \in [n]$ consider the function $p_i : \{0,1\}^m \to \mathbb{R}_{\geq 0}$ defined by

$$p_i(x) := \ell_i(\mu^{\mathrm{GapOr}_m(x)}).$$

First note that p_i can always be written as a conical combination of q_1 and q_2 in degree $\log n$. Moreover, if ℓ_i is $(\epsilon, \mu, 0)$-noisy, meaning $\ell_i(\mu^1) \geq \epsilon \cdot \ell_i(\mu^0)$, then we can do better and write p_i as a conical combination of q_1 and q_3 in degree $\log(1/\epsilon)$. We now define $p_\ell := \prod_i p_i$. The claimed bound on the degree of p_ℓ follows because at most δ fraction of the ℓ_i are non-noisy. □

We are now ready to prove Theorem3 using Multileaf Lemma and Simulation Lemma.

Proof of Theorem3. Suppose for contradiction that T is a randomised decision tree for $f \circ g^n$ having error $1/3$ and depth $\gamma n \overline{R}_{1/n}(g)$ where $\gamma = o(1)$ as $n \to \infty$. Our goal is to construct an $o(n \log n)$-degree $o(1)$-approximate conical junta for $f \circ \mathrm{GapOr}^n_{\log n}$ (or $f \circ \neg\mathrm{GapOr}^n_{\log n}$). We make two simplifying assumptions wlog.

1. The randomised tree T has error $o(1)$. To ensure this, we may reduce T's error by running it $1/\sqrt{\gamma} = \omega(1)$ times. This will yield an $o(1)$-error tree of depth $\sqrt{\gamma} n \overline{R}_{1/n}(g) = o(n \overline{R}_{1/n}(g))$.

2. There is some $\epsilon := 1/n^{o(1)}$ such that T has depth $o(n\overline{R}_\epsilon(g))$. To ensure this, we may apply Claim7 to see that $\gamma n\overline{R}_{1/n}(f) \le \sqrt{\gamma}n\overline{R}_\epsilon(f) \le o(n\overline{R}_\epsilon(f))$ where $\epsilon := 1/n^{4\sqrt{\gamma}}$.

We invoke Multileaf Lemma with the above $\epsilon \le o(1)$ and $\delta := \max\{\gamma^c, \epsilon^c\} \le o(1)$ for small enough constant $c > 0$. We get a hard distribution μ and a hard side b, say $b = 0$ (case $b = 1$ is similar, but using $\neg$GapOr), such that the following holds: For every t in the support of T if we run t on a random input $x \sim \mu^y$, where $y \in \{0,1\}^n$, then the leaf reached ℓ_x^t will be $(\delta, \epsilon, \mu, 0)$-noisy with probability $1 - o(1)$. This allows us to effectively ignore non-noisy leaves: denoting by $N(t) \subseteq A(t)$ the set of accepting leaves that are $(\delta, \epsilon, \mu, 0)$-noisy, we have

$$\begin{aligned} \forall y \in \{0,1\}^n : \quad \mathbb{E}_{x\sim\mu^y}[t(x)] &= \mathbb{E}_{x\sim\mu^y}\left[\sum_{\ell\in A(t)} \ell(x)\right] && \text{(Using(1))} \\ &\in \mathbb{E}_{x\sim\mu^y}\left[\sum_{\ell\in N(t)} \ell(x)\right] \pm o(1). \end{aligned} \tag{2}$$

We now define the approximating conical junta by

$$p(x) := \mathbb{E}_{t\sim T}\left[\sum_{\ell\in N(t)} p_\ell(x)\right],$$

where the p_ℓ are given by Simulation Lemma. Hence p has degree at most

$$n \cdot [\delta \log n + \log(1/\epsilon)] = n \cdot [o(1)\log n + \log n^{o(1)}] = o(n \log n).$$

We finish the proof of Theorem3 by verifying that p indeed $o(1)$-approximates $f \circ \mathrm{GapOr}^n_{\log n}$.

$$\begin{aligned} \forall x : \quad p(x) &= \mathbb{E}_{t\sim T}\left[\sum_{\ell\in N(t)} p_\ell(x)\right] && \left(y := \mathrm{GapOr}^n_{\log n}(x)\right) \\ &= \mathbb{E}_{t\sim T}\left[\sum_{\ell\in N(t)} \ell(\mu^y)\right] \\ &= \mathbb{E}_{t\sim T}\left[\sum_{\ell\in N(t)} \mathbb{E}_{x'\sim\mu^y}[\ell(x')]\right] \\ &= \mathbb{E}_{t\sim T}\left[\mathbb{E}_{x'\sim\mu^y}\left[\sum_{\ell\in N(t)} \ell(x')\right]\right] \\ &\in \mathbb{E}_{t\sim T}\left[\mathbb{E}_{x'\sim\mu^y}[t(x')]\right] \pm o(1) && \text{(Using(2))} \\ &= \mathbb{E}_{x'\sim\mu^y}\left[\mathbb{E}_{t\sim T}[t(x')]\right] \pm o(1) \\ &\in \mathbb{E}_{x'\sim\mu^y}[(f\circ g^n)(x')] \pm o(1) && (T \text{ has error } o(1)) \\ &= f(y) \pm o(1) = (f \circ \mathrm{GapOr}^n_{\log n})(x) \pm o(1). \end{aligned}$$

□

4 Proof of Leaf Lemma

We prove Leaf Lemma in three subsections. (Section4.1) We start by recalling a distributional characterisation due to Blais and Brody [7] of expected query complexity $\overline{R}_\epsilon$ using decision trees that can "abort". (Section4.2) We then formulate a Hard Side Lemma, which encapsulates the core challenge in finding the hard side of a given function g and from which Leaf Lemma is easy to derive. (Section4.3) Finally, we prove the Hard Side Lemma.

4.1 Distributional characterisation of $\overline{R}_\epsilon$ due to Blais–Brody

A (deterministic) abort-tree t is a decision tree that outputs either a boolean value (0 or 1) or the abort symbol $\perp$. When an abort-tree is trying to compute a boolean function g, we do not consider the output $\perp$ as an "error"; the tree simply gives up on the computation. Indeed, we say that $t(x)$ errs iff $t(x) = 1 - g(x)$, that is, $t(x) \neq \perp$ and $t(x) \neq g(x)$. As before, a randomised abort-tree is a probability distribution over deterministic abort-trees. For $\gamma \in (0,1)$ and $\epsilon \in [0, 1/2)$ we define $R_{\gamma,\epsilon}(g)$ as the least (worst-case) depth of a randomised abort-tree T such that for all $x \in \mathrm{dom}(g)$:

$$\Pr_{t\sim T}[t(x) = \perp] \leq \gamma \quad \text{and} \quad \Pr_{t\sim T}[t(x) \text{ errs}] \leq \epsilon.$$

We formulate a distributional version of $R_{\gamma,\epsilon}(g)$ as follows. For a distribution μ over $\mathrm{dom}(g)$, we define $D^\mu_{\gamma,\epsilon}(g)$ as the least depth of a deterministic abort-tree t such that

$$\Pr_{x\sim\mu}[t(x) = \perp] \leq \gamma \quad \text{and} \quad \Pr_{x\sim\mu}[t(x) \text{ errs}] \leq \epsilon.$$

The following two lemmas from [7, Section3.1] connect abort-trees and $\overline{R}_\epsilon(g)$.

Lemma8 (Abort vs. expected depth). For every $\epsilon \in [0, 1/2)$ and $\gamma \in (0,1)$,

$$\gamma \cdot R_{\gamma,\epsilon}(g) \leq \overline{R}_\epsilon(g) \leq \frac{1}{1-\gamma} \cdot R_{\gamma,(1-\gamma)\epsilon}(g).$$

Lemma9 (Minimax). For every $\epsilon \in [0, 1/2)$, $\gamma \in (0,1)$, and $\alpha, \beta \in (0,1)$ with $\alpha + \beta \leq 1$,

$$\max_\mu D^\mu_{\gamma/\alpha,\epsilon/\beta}(g) \leq R_{\gamma,\epsilon}(g) \leq \max_\mu D^\mu_{\alpha\gamma,\beta\epsilon}(g).$$

4.2 Statement of Hard Side Lemma

When searching for the hard side of a partial function g under a distribution $\mu = \frac{1}{2}(\mu^0 + \mu^1)$, it is convenient to study a more symmetric notion of noisiness than the one-sided variant defined earlier. For a leaf $\ell \in L(t)$ of an abort-tree t, we define the relative error $\mathrm{re}(\ell, \mu)$ so that if ℓ is an aborting leaf, then $\mathrm{re}(\ell, \mu) := 0$; otherwise

$$\mathrm{re}(\ell, \mu) := \frac{\min\{\ell(\mu^0), \ell(\mu^1)\}}{\ell(\mu^0) + \ell(\mu^1)} \in [0, 1/2].$$

This definition captures the best achievable error of a leaf in an abort-tree. Namely, let us say that t is μ-smart if every non-abort leaf $\ell \in L(t)$ outputs a boolean value $i \in \{0,1\}$ that maximises $\ell(\mu^i)$. Then for every leaf ℓ in a μ-smart t we have $\Pr_{x\sim\mu}[t(x) \text{ errs} \mid \ell_x^t = \ell] = \mathrm{re}(\ell,\mu)$. An easy calculation gives the following claim, which we record for future use.

Claim10. For a μ-smart t we have $\Pr_{x\sim\mu}[t(x) \text{ errs}] = \mathbb{E}_{x\sim\mu}[\mathrm{re}(\ell_x^t,\mu)]$.

Another easy calculation shows that relative error implies noisiness.

Claim11. If $\mathrm{re}(\ell,\mu) \geq \epsilon$, then ℓ is (ϵ,μ,b)-noisy for both $b \in \{0,1\}$.

We are now ready to formulate Hard Side Lemma, which isolates the technical challenge 3.1 (discussed in Subsection3.1): Every partial function g admits a balanced distribution μ and a hard side b such that if we run a shallow abort-tree on the hard side μ^b of μ, then t must either abort with high probability or we reach a leaf of noticeable error (in expectation).

Hard Side Lemma. For every partial function g and $0 < \epsilon \lll \delta \ll 1$, there exists a distribution $\mu = \frac{1}{2}(\mu^0+\mu^1)$ over $\mathrm{dom}(g)$ and a hard side $b \in \{0,1\}$ such that for any deterministic abort-tree t with $\mathrm{depth}(t)/\overline{R}_\epsilon(g) \lll \delta$ we have either

$$\Pr_{x\sim\mu^b}[t(x) = \perp] > 1-\delta \quad \text{or} \quad \mathbb{E}_{x\sim\mu^b}[\mathrm{re}(\ell_x^t,\mu)] > \epsilon. \tag{3}$$

We defer the proof until Subsection4.3. We first use the lemma to prove Leaf Lemma, and here is where we address challenge 3.1: we exploit the high abort probability (namely, $1-\delta$) guaranteed by Hard Side Lemma to show that typical leaves are noisy.

Leaf Lemma (restated). For every partial g and $0 < \epsilon \lll \delta \ll 1$, there exists a distribution $\mu = \frac{1}{2}(\mu^0+\mu^1)$ over $\mathrm{dom}(g)$ and a hard side $b \in \{0,1\}$ such that for every deterministic tree t and $i \in \{0,1\}$:

$$\frac{\mathbb{E}_{x\sim\mu^i}[q(t,x)]}{\overline{R}_\epsilon(g)} \lll \delta \quad \Longrightarrow \quad \Pr_{x\sim\mu^i}[\ell_x^t \text{ is } (\epsilon,\mu,b)\text{-noisy}] \geq 1-\delta.$$

Proof. We observe first that regardless of μ, b, or even the expected depth of t, the lemma holds for the easy side $i = 1-b$. Indeed, if we let $B \subseteq L(t)$ denote the set of non-(ϵ,μ,b)-noisy leaves,

$$\Pr_{x\sim\mu^{1-b}}[\ell_x^t \in B] = \sum_{\ell\in B} \ell(\mu^{1-b}) < \epsilon \sum_{\ell\in B} \ell(\mu^b) \leq \epsilon \sum_{\ell\in L(t)} \ell(\mu^b) = \epsilon \leq \delta.$$

Let us then focus on the interesting case $i = b$ where the careful choice of μ and b is essential. We invoke Hard Side Lemma with parameters ϵ and $\dot\delta := \delta^2$ (assuming suitably $0 < \epsilon \lll \dot\delta \ll 1$) to obtain μ and b such that for every abort-tree $\dot t$ with $\mathrm{depth}(\dot t) \leq \dot\delta^c \overline{R}_\epsilon(g)$ the property(3) holds (with dotted parameters). Let $x \sim \mu^b$ henceforth and write $\mathrm{re}(\ell) := \mathrm{re}(\ell,\mu)$ for short. Suppose t satisfies $\mathbb{E}_x[q(t,x)] \leq \dot\delta^{c+2}\overline{R}_\epsilon(g)$ (where we chose $2(c+2)$ as the exponent hidden by $\lll$). Recalling from Claim11 that relative error implies noisiness, our goal is to show

$$\Pr_x[\mathrm{re}(\ell_x^t) \geq \epsilon] \geq 1-\delta. \tag{4}$$

We convert t into an abort-tree by letting t' be a modification of t that aborts whenever more than $\dot{\delta}^c \overline{R}_\epsilon(g)$ queries are made. Using Markov's inequality and the low expected depth of t,

$$\Pr_x[t'(x) = \perp] = \Pr_x[q(t,x) > \dot{\delta}^c \overline{R}_\epsilon(g)] \leq \frac{\mathbb{E}_x[q(t,x)]}{\dot{\delta}^c \overline{R}_\epsilon(g)} \leq \dot{\delta}^2.$$

We also have $\Pr_x[\mathrm{re}(\ell_x^t) \geq \epsilon] \geq \Pr_x[\mathrm{re}(\ell_x^{t'}) \geq \epsilon]$ since we only made more executions abort. To prove(4), suppose for contradiction that $\Pr_x[\mathrm{re}(\ell_x^{t'}) \geq \epsilon] < 1 - \delta$. Let $\dot{t}$ be a further modification of t' that aborts any leaf $\ell \in L(t')$ with $\mathrm{re}(\ell) \geq \epsilon$. Note that

$$\Pr_x[\dot{t}(x) = \perp] \leq \Pr_x[t'(x) = \perp] + \Pr_x[\mathrm{re}(\ell_x^{t'}) \geq \epsilon] \leq \dot{\delta}^2 + 1 - \delta \leq 1 - \dot{\delta}.$$

Hence we get from (dotted) property(3) that $\mathbb{E}_x[\mathrm{re}(\ell_x^{\dot{t}})] > \epsilon$. But this contradicts the fact that $\mathrm{re}(\ell) < \epsilon$ for all $\ell \in L(\dot{t})$ by construction. This verifies(4) and concludes the proof. □

4.3 Proof of Hard Side Lemma

Let ν be a distribution that witnesses $D := \max_{\nu'} D^{\nu'}_{1-\delta,\epsilon^{1/3}}(g)$ so that every abort-tree t with $\mathrm{depth}(t) < D$ fails to satisfy at least one of the following:

$$\Pr_{x\sim\nu}[t(x) = \perp] \leq 1 - \delta, \tag{5}$$

$$\Pr_{x\sim\nu}[t(x) \text{ errs}] \leq \epsilon^{1/3}. \tag{6}$$

As a minor technicality, we re-balance ν. We can write $\nu = \lambda\mu^0 + (1-\lambda)\mu^1$ where $\lambda \in (0,1)$ and μ^i is a distribution supported on $g^{-1}(i)$. We define $\mu := \frac{1}{2}(\mu^0 + \mu^1)$ as our balanced distribution. Assume towards a contradiction that there does not exist a hard side for μ, that is, the claim of the lemma fails for both $b \in \{0,1\}$. This means there exists two abort-trees t_0 and t_1 of depth at most $\delta^3 \overline{R}_\epsilon(g)$ (where we chose 3 as the exponent hidden by $\lll$) such that for both $b \in \{0,1\}$:

$$\Pr_{x\sim\mu^b}[t_b(x) = \perp] \leq 1 - \delta, \tag{7}$$

$$\mathbb{E}_{x\sim\mu^b}[\mathrm{re}(\ell_x^{t_b}, \mu)] \leq \epsilon. \tag{8}$$

We will use t_0 and t_1 to construct a third tree t that computes g too well relative to ν contradicting our choice of ν. We may assume wlog that t_0 and t_1 are μ-smart, since the properties(7)–(8) do not depend on the boolean leaf-labels (only whether a leaf aborts or not). We now define t as follows: On input x we run both $t_0(x)$ and $t_1(x)$; if $t_0(x) \neq \perp$, we output $t_0(x)$; otherwise we output $t_1(x)$. We will show that t has $\mathrm{depth}(t) < D$ and satisfies(5)–(6), which will contradict our choice of ν.

Tree t is shallow. We have the following chain of inequalities

$$\begin{aligned}\mathrm{depth}(t) &\leq 2\delta^3 \overline{R}_\epsilon(g) \leq 32\delta^3 \overline{R}_{\epsilon^{1/4}}(g) < \delta^2 \overline{R}_{\epsilon^{1/4}}(g) \leq R_{1-\delta^2,\delta^2\epsilon^{1/4}}(g) \\ &\leq \max_{\nu'} D^{\nu'}_{(1-\delta^2)^2,\delta^4\epsilon^{1/4}}(g) \leq \max_{\nu'} D^{\nu'}_{1-\delta,\epsilon^{1/3}}(g) =: D.\end{aligned}$$

The first inequality uses the definition of t. Second uses error reduction (Claim7 with $k := 4$). Third uses $\delta \ll 1$. Fourth uses Lemma8 (with $\gamma := 1 - \delta^2$). Fifth uses the minimax lemma (Lemma9 with $\alpha := 1 - \delta^2$, $\beta := \delta^2$). The final inequality uses $\epsilon \lll \delta \ll 1$.

Tree t has bounded abort. We verify property(5) by

$$\begin{aligned}\Pr_{x\sim\nu}[t(x) = \perp] &= \Pr_{x\sim\nu}[t_0(x) = \perp \wedge t_1(x) = \perp] \\ &= \lambda \Pr_{x\sim\mu^0}[t_0(x) = \perp \wedge t_1(x) = \perp] + (1-\lambda) \Pr_{x\sim\mu^1}[t_0(x) = \perp \wedge t_1(x) = \perp] \\ &\leq \lambda \Pr_{x\sim\mu^0}[t_0(x) = \perp] + (1-\lambda) \Pr_{x\sim\mu^1}[t_1(x) = \perp] \leq 1 - \delta. \quad \text{(Using(7))}\end{aligned}$$

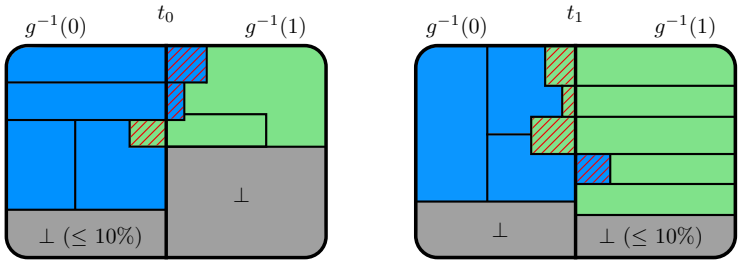


Figure1. Two trees t_0 and t_1 in the proof of Hard Side Lemma. The leaves partition $\mathrm{dom}(g)$ into subcubes where grey leaves output $\perp$, green leaves output 1, and blue leaves output 0. Hatched regions are error. We are promised that, e.g., t_0 has bounded abort (10% in our figure) over μ^0, but not necessarily over μ^1.

Tree t errs rarely. We start with a claim that says that if the expected relative error is low over one side μ^b of μ, then a μ-smart tree errs rarely over the whole distribution μ.

Claim12. Let t' be μ-smart and $b \in \{0,1\}$. If $\mathbb{E}_{x\sim\mu^b}[\mathrm{re}(\ell_x^{t'}, \mu)] \leq \epsilon$ then $\Pr_{x\sim\mu}[t'(x) \text{ errs}] \leq \epsilon^{1/2}$.

Proof. We prove the claim for $b = 0$ as the other case is analogous. Since t' and μ are fixed, we drop them from notation writing $\mathrm{re}(\ell) := \mathrm{re}(\ell, \mu)$, $\ell_x := \ell_x^{t'}$, $L := L(t')$. We argue that relative error on one side of the distribution must spill over to the other side:

$$\begin{aligned}\mathbb{E}_{x\sim\mu^0}[\mathrm{re}(\ell_x)] &= \sum_{\ell\in L} \ell(\mu^0)\,\mathrm{re}(\ell) \geq \sum_{\ell\in L} \ell(\mu^1)\,\mathrm{re}(\ell)^2 \\ &= \mathbb{E}_{x\sim\mu^1}[\mathrm{re}(\ell_x)^2] \geq \mathbb{E}_{x\sim\mu^1}[\mathrm{re}(\ell_x)]^2.\end{aligned}$$

Here the first inequality used $\ell(\mu^0) \geq \ell(\mu^1)\,\mathrm{re}(\ell)$ (from Claim11) and the second inequality used Jensen's inequality. It follows that $\mathbb{E}_{x\sim\mu^1}[\mathrm{re}(\ell_x)] \leq \mathbb{E}_{x\sim\mu^0}[\mathrm{re}(\ell_x)]^{1/2} \leq \epsilon^{1/2}$ and therefore $\mathbb{E}_{x\sim\mu}[\mathrm{re}(\ell_x)] \leq \epsilon^{1/2}$. The claim then follows from Claim10. □

We now verify property(6), which concludes the proof of Hard Side Lemma.

$$\begin{aligned}
\Pr_{x\sim\nu}[t(x) \text{ errs}] &\le \Pr_{x\sim\nu}[t_0(x) \text{ errs} \vee t_1(x) \text{ errs}] \\
&\le \sum_{b\in\{0,1\}} \Pr_{x\sim\nu}[t_b(x) \text{ errs}] \\
&= \sum_{b\in\{0,1\}} \lambda \Pr_{x\sim\mu^0}[t_b(x) \text{ errs}] + (1-\lambda) \Pr_{x\sim\mu^1}[t_b(x) \text{ errs}] \\
&\le \sum_{b\in\{0,1\}} 2 \Pr_{x\sim\mu}[t_b(x) \text{ errs}] \le \sum_{b\in\{0,1\}} 2\cdot\epsilon^{1/2} \quad \text{(Claim12 and(8))} \\
&= 4\epsilon^{1/2} \le \epsilon^{1/3}. \quad (\epsilon \ll 1)
\end{aligned}$$

5 Proof of Multileaf Lemma

Multileaf Lemma (restated). For every partial g and $0 < \epsilon \lll \delta \ll 1$, there exists a distribution $\mu = \frac{1}{2}(\mu^0 + \mu^1)$ over $\mathrm{dom}(g)$ and a hard side $b \in \{0,1\}$ such that for every deterministic tree t taking inputs from $\mathrm{dom}(g^n)$ and having $\mathrm{depth}(t)/(n\overline{R}_\epsilon(g)) \lll \delta$,

$$\forall y \in \{0,1\}^n : \quad \Pr_{x\sim\mu^y}[\ell_x^t \text{ is } (\delta,\epsilon,\mu,b)\text{-noisy}] \ge 1-\delta.$$

Proof. Apply Leaf Lemma with parameters ϵ and $\dot\delta := \delta^3$ (assuming suitably $0 < \epsilon \lll \dot\delta \ll 1$) to obtain $\mu = \frac{1}{2}(\mu^0+\mu^1)$ and $b \in \{0,1\}$ that satisfy the lemma for trees of depth at most $\dot\delta^c \overline{R}_\epsilon(g)$. Fix $y \in \{0,1\}^n$ and a deterministic tree t over $\mathrm{dom}(g^n)$ with $\mathrm{depth}(t) \le \dot\delta^{c+4} n\overline{R}_\epsilon(g)$ (where we chose $3(c+4)$ as the exponent hidden by $\lll$). Here is the plan for our proof. An input $x \in \mathrm{dom}(g^n)$ can be seen as inducing several subtrees of t corresponding to distinct coordinates $i \in [n]$. Indeed, define $t_{x,i}$ as the tree over inputs from $\mathrm{dom}(g)$ that is obtained from t by substituting x as its input variables except retaining x_i as free variables. If we can show that $t_{x,i}$ has shallow depth in expectation over an input $z \sim \mu^{y_i}$ then we can hope to use and argue that the reached leaf $\ell_z \in L(t_{x,i})$ (which is one of the n components of a leaf of t) is typically (ϵ,μ,b)-noisy. Let us formalise this plan. Let $x \sim \mu^y$ henceforth. For $i \in [n]$ we define two events

$$\begin{aligned}
i\text{-th tree is shallow:} &\quad S_i(x) \iff \mathbb{E}_{z\sim\mu^{y_i}}[q(t_{x,i},z)] \le \dot\delta^c \overline{R}_\epsilon(g), \\
i\text{-th leaf is noisy:} &\quad N_i(x) \iff \ell_{x_i} \in L(t_{x,i}) \text{ is } (\epsilon,\mu,b)\text{-noisy}.
\end{aligned}$$

Note that Leaf Lemma states $\Pr_x[N_i \mid S_i] \ge 1-\dot\delta$. Thinking of S_i and N_i as indicator variables, we define $S := \frac{1}{n}\sum_i S_i$ and $N := \frac{1}{n}\sum_i N_i$. With this notation, Multileaf Lemma becomes equivalent to

$$\Pr_x[N \ge 1-\delta] \ge 1-\delta. \tag{9}$$

To show this, we compute as follows (using Claim13 that is proved below)

$$\mathbb{E}_x[N] = \frac{1}{n}\sum_i \Pr_x[N_i] \ge \frac{1}{n}\sum_i (1-\dot\delta)\Pr[S_i] \quad \text{(Leaf Lemma)}$$

$$= (1-\dot{\delta})\mathbb{E}_x[S] \geq (1-\dot{\delta})(1-\dot{\delta}) \quad \text{(Claim13)}$$
$$\geq 1-\delta^2. \quad (\dot{\delta} := \delta^3 \ll 1)$$

Hence(9) follows by applying Markov's inequality to the nonnegative random variable $1-N \geq 0$. This completes the proof apart from the following claim. □

Claim13. $\mathbb{E}_x[S] \geq 1-\dot{\delta}$.

Proof. Let $q_i(t,x)$ denote the number of queries made by t to the i-th component of x. Define $x^{i\leftarrow z}$ as a copy of x but where z is inserted at the i-th component. Note that $q_i(t, x^{i\leftarrow z}) = q(t_{x,i}, z)$. Linearity of expectation gives

$$\sum_{i\in[n]} \mathbb{E}_x[q_i(t,x)] \leq \text{depth}(t) \leq \dot{\delta}^{c+4} n \overline{R}_\epsilon(g). \tag{10}$$

Define $I \subseteq [n]$ as the set of coordinates i satisfying

$$\mathbb{E}_x[q_i(t,x)] \leq \dot{\delta}^{c+2}\overline{R}_\epsilon(g). \tag{11}$$

We have that $|I| \geq (1-\dot{\delta}^2)n$ as otherwise more than $\dot{\delta}^2 n$ terms in the sum(10) are larger than $\dot{\delta}^{c+2}\overline{R}_\epsilon(g)$ contradicting the upper bound on depth(t). Fix $i \in I$. Sampling $x \sim \mu^y$ is equivalent to first taking $x \sim \mu^y$, then sampling independently $z \sim \mu^{y_i}$, and finally outputting $x^{i\leftarrow z}$. Hence

$$\mathbb{E}_x \mathbb{E}_{z\sim\mu^{y_i}}[q_i(t, x^{i\leftarrow z})] = \mathbb{E}_x[q_i(t,x)] \leq \dot{\delta}^{c+2}\overline{R}_\epsilon(g).$$

We get from Markov's inequality and the above that

$$\Pr_x[\neg S_i] = \Pr_x\left[\mathbb{E}_{z\sim\mu^{y_i}}[q_i(t, x^{i\leftarrow z})] > \dot{\delta}^c \overline{R}_\epsilon(g)\right] \leq \dot{\delta}^2. \tag{12}$$

In conclusion,

$$\mathbb{E}_x[S] \geq \frac{1}{n}\sum_{i\in I} \Pr_x[S_i] \geq \frac{1}{n}|I| \cdot (1-\dot{\delta}^2) \geq (1-\dot{\delta}^2)^2 \geq 1-\dot{\delta}.$$

□

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